

Chapter Summary

Key Terms

11.1 Voting Methods

- majority
- plurality
- runoff election
- runoff voting system
- two-round system
- Hare Method
- preference ranking
- ranked ballot
- ranked-choice voting (RCV)
- instant runoff voting (IRV)
- Borda count method
- Borda score
- Compromise candidate
- divisive candidate
- pairwise comparison method
- Condorcet voting methods
- Condorcet candidate
- approval voting system
- approval voting ballot
- spoiler

11.2 Fairness in Voting Methods

- majority criterion
- tyranny of the majority
- Condorcet criterion
- Condorcet method
- monotonicity criterion
- up-rank
- down-rank
- independence of irrelevant alternatives criterion (IIA)
- Arrow's Impossibility Theorem
- cardinal voting system

11.3 Standard Divisors, Standard Quotas, and the Apportionment Problem

- apportion
- apportionment problem
- proportional
- part-to-part ratio
- representative democracies
- standard divisor
- states
- seats
- house size
- state population
- total population
- standard quota

11.4 Apportionment Methods

- apportionment problem
- lower quota
- upper quota

11.5 Fairness in Apportionment Methods

- apportionment paradox
- Alabama paradox
- reapportionment
- population growth rate
- population paradox
- new-state paradox

Key Concepts

11.1 Voting Methods

- In plurality voting, the candidate with the most votes wins.
- When a voting method does not result in a winner, runoff voting can be used to do so.
- Ranked-choice voting, also known as instant runoff voting, is one type of ranked voting system.
- The Borda count method is a type of ranked voting system in which each candidate is given a Borda score based on the number of candidates ranked lower than them on each ballot.
- When pairwise comparison is used, the winner will be the Condorcet candidate if one exists.
- Approval voting allows voters to give equally weighted votes to multiple candidates.
- When a voter finds a characteristic of a particular voting method unappealing, they may consider that characteristic a flaw in the voting method and look for an alternative method that does not have that characteristic.

11.2 Fairness in Voting Methods

- There are several common measures of voting fairness, including the majority criterion, the head-to head criterion, the monotonicity criterion, and the irrelevant alternatives criterion.
- According to Arrow's Impossibility Theorem, each voting method in which the only information is the order of preference of the voters will violate one of the fairness criteria.

11.3 Standard Divisors, Standard Quotas, and the Apportionment Problem

- The apportionment problem is how to fairly divide and distribute available resources to recipients in whole, not fractional, parts.
- To distribute the seats in the U.S. House of Representatives fairly to each state, calculations are based on state population, total population, and house size, or the total number of seats to be apportioned.
- The standard divisor is the ratio of the total population to the house size, and the standard quota is the number of seats that each state should receive.

11.4 Apportionment Methods

- Hamilton's method of apportionment uses the standard divisor and standard lower quotas, and it distributes any remaining seats based on the size of the fractional parts of the standard lower quota. Hamilton's method satisfies the quota rule and favors neither larger nor smaller states.
- Jefferson's method of apportionment uses a modified divisor that is adjusted so that the modified lower quotas sum to the house size. Jefferson's method violates the quota rule and favors larger states.
- Adams's method of apportionment uses a modified divisor that is adjusted so that the modified upper quotas, sum to the house size. Adams's method violates the quota rule and favors smaller states.
- Webster's method of apportionment uses a modified divisor that is adjusted so that the modified state quotas, rounded using traditional rounding, sum to the house size. Webster's method violates the quota rule but favors neither larger nor smaller states.

11.5 Fairness in Apportionment Methods

- Several surprising outcomes can occur when apportioning seats that voters may find unfair: Alabama paradox, population paradox, and new-state paradox.
- Apportionment methods are susceptible to apportionment paradoxes and may violate the quota rule.
- The Balinsky-Young Impossibility Theorem indicates that no apportionment can satisfy all fairness criteria.

Videos

11.1 Voting Methods

- [How Does Ranked-Choice Voting Work? \(https://openstax.org/r/ranked-choice_voting1\)](https://openstax.org/r/ranked-choice_voting1)
- [Determine Winner of Election by Ranked-Choice Method \(aka Instant Runoff\) \(https://openstax.org/r/ranked-determine-winner\)](https://openstax.org/r/ranked-determine-winner)

[choice_voting2\)](#)

- [Determine Winner of Election by Borda Count Method \(https://openstax.org/r/Borda_count_method\)](https://openstax.org/r/Borda_count_method)
- [Determine Winner of Election by Pairwise Comparison Method \(https://openstax.org/r/w1NNK7Dn3E8\)](https://openstax.org/r/w1NNK7Dn3E8)

11.2 Fairness in Voting Methods

- [Separation of Powers and Checks and Balances \(https://openstax.org/r/separation_of_powers\)](https://openstax.org/r/separation_of_powers)

11.3 Standard Divisors, Standard Quotas, and the Apportionment Problem

- [What Is a Ratio? \(https://openstax.org/r/ratios_proportions\)](https://openstax.org/r/ratios_proportions)
- [What Are the Different Types of Ratios? \(https://openstax.org/r/ratio_types\)](https://openstax.org/r/ratio_types)
- [Math Antics – Rounding \(https://openstax.org/r/rounding\)](https://openstax.org/r/rounding)

11.4 Apportionment Methods

- [Hamilton Method of Apportionment \(https://openstax.org/r/Hamiltons_method\)](https://openstax.org/r/Hamiltons_method)
- [Jefferson Apportionment Method \(https://openstax.org/r/Jeffersons_method\)](https://openstax.org/r/Jeffersons_method)
- [Adams Method Apportionment Calculator \(https://openstax.org/r/Adams_method\)](https://openstax.org/r/Adams_method)

Formula Review

11.3 Standard Divisors, Standard Quotas, and the Apportionment Problem

Let A be a particular item and B another such that there is a constant ratio of A to B .

- ratio of B 's to A 's = $\frac{1}{\text{ratio of } A\text{'s to } B\text{'s}}$ and ratio of A 's to B 's = $\frac{1}{\text{ratio of } B\text{'s to } A\text{'s}}$
- units of A = (units of B) $\times$ (ratio of A 's to B 's) = $\frac{\text{units of } B}{\text{ratio of } B\text{'s to } A\text{'s}}$
- units of B = (units of A) $\times$ (ratio of B 's to A 's) = $\frac{\text{units of } A}{\text{ratio of } A\text{'s to } B\text{'s}}$

$$\text{Standard Divisor} = \frac{\text{Total Population}}{\text{House Size}}$$

$$\text{State's Standard Quota} = \frac{\text{State Population}}{\text{Standard Divisor}} \text{ seats}$$

11.5 Fairness in Apportionment Methods

$$\text{population growth rate} = \frac{\text{current population size} - \text{previous population size}}{\text{previous population size}}$$

$$\text{New House Size} = \frac{\text{New Population}}{\text{Original Standard Divisor}} \text{ rounded to the nearest whole number.}$$

Projects

The First Census: Challenges of Collecting Population Data

As you and your fellow founders write the Imaginarian constitution, you must also design systems to collect accurate information about the population of Imaginaria. Why is this important and what are the challenges you will face? Let's find out! Research and answer each question. (Questions adapted from "Authorizing the First Census—The Significance of Population Data," Census.gov, United States Census Bureau)

1. The First Congress laid out a plan for collecting this data in Chapter II, *An Act Providing for the Enumeration of the Inhabitants of the United States*, which was approved March 1, 1790. What group did Congress select to carry out the first enumeration? Why did they choose this group? What might be the advantages and disadvantages to this approach?
2. Choose at least two challenges faced by the U.S. government during the first enumeration and explain how the information gathered might have helped address them.
3. President James Madison was a U.S. Representative who participated in the census debate in the first congress. What were the main points of his remarks and how were they relevant to the overall debate about the first enumeration?
4. What issues do you think the U.S. Census Bureau encounters today as it continues to collect and process data about the U.S. population that might be significant to you and the other founders of Imaginaria?

The Party System: How Many Political Parties Are Enough?

The electoral system of Imaginaria will likely involve multiple political parties. The way these parties interact with the system may be determined by the founders of the new democracy. Let's explore the ways in which political parties interact with the governments of various democracies by researching and answering each of the following questions.

1. What concerns did the founders of the United States have about political parties? Have any of their concerns become a reality? Were political parties addressed in the U.S. Constitution? How did they become such a large part of politics in the U.S.? As a founder of Imaginaria, would you address political parties in your constitution?
2. What is frontloading? Does our current system of frontloading impact fair representation? Why do two small, racially homogeneous states hold their primaries first? Do you think this impacts the final results?
3. How does the interaction political parties and the electoral system in the United States differ from that of other countries? Give at least three specific examples. Of these three, which would you be most likely to use as a model for Imaginaria?

Technology and Voting: How Has the Digital Age Impacted Elections?

Unlike most other countries, Imaginaria will be founded in the digital age. Let's explore the impact this might have on how you choose to set up your electoral system. Research and answer each question.

1. Participation in an electoral system is very important. In what ways does the Internet positively affect participation? In what ways does it negatively affect participation. What role, if any, should government of Imaginaria play in tempering or promoting the impact of the Internet?
2. Find at least three examples of governments that utilize Internet voting around the world. What concerns have slowed the spread of this technology? What are the advantages and disadvantages of Internet voting? Would you be in favor of Internet voting for Imaginaria? Why or why not?

Chapter Review

Voting Methods

1. In a plurality election, the candidates have the following vote counts: A 125, B 132, C 149, and D 112. Which candidate has the plurality and wins the election?

For the following exercises, use the table below.

Options	A	B	C	D	E
Candidate 1	1	3	3	1	3
Candidate 2	2	1	1	2	4
Candidate 3	3	4	2	4	1
Candidate 4	4	2	4	3	2

2. Which candidate has a plurality?
3. Does the plurality candidate have a majority?
4. Determine the winner of the election by the Hare method based on the sample preference summary in the table.

For the following exercises, use the table below.

Number of Ballots	10	20	15	5
Option A	1	4	3	4
Option B	2	3	4	2
Option C	4	2	1	3
Option D	3	1	2	1

5. Use ranked-choice voting to determine the two options in the final round and the number of votes they each receive in that round.
6. Is there a winning option? If so, which option? Justify your answer.

For the following exercises, use the table below.

Number of Ballots	100	80	110	105	55
Candidate A	1	1	4	4	2
Candidate B	2	2	2	3	1
Candidate C	4	4	1	1	4
Candidate D	3	3	3	2	3

7. What are the Borda scores for each candidate?
8. Which candidate is the winner by the Borda count method?

Use Pairwise Comparison Matrix for Candidates U, V, W, X, and Y to answer Questions 9 and 10.

9. Calculate the points received by each candidate in pairwise comparison matrix.
10. Determine the winner of the pairwise comparison election represented by matrix. If there is a winner,

determine whether the winner is a Condorcet candidate and explain your reasoning. If there is no winner, indicate this.

11. The ladies of *The Big Bang Theory* decide to hold their own approval voting election to determine the best option in Rock, Paper, Scissors, Lizard, Spock. Use the summary of their approval ballots in the table below to determine the number of votes for each candidate. Determine the winner, or state that there is none.

VOTERS	Penny	Bernadette	Amy
Rock	Yes	No	No
Paper	Yes	Yes	No
Scissors	Yes	Yes	Yes
Lizard	No	No	No
Spock	Yes	No	Yes

For the following exercises, use the table below.

Percentage of Vote	40%	35%	25%
Candidate A	1	3	2
Candidate B	2	1	3
Candidate C	3	2	1

12. Which candidate is the winner by the ranked-choice method?
 13. Suppose that you used the approval method and each voter approved their top two choices. Which candidate is the winner by the approval method?
 14. Which candidate is the winner by the Borda count method?

Fairness in Voting Methods

15. In a Borda count election, the candidates have the following Borda scores: A 1245, B 1360, and C 787. Candidate A received 55% of the first place rankings. Identify which fairness criteria, if any, are violated by characteristics of the described voter profile in this Borda count election. Explain your reasoning.

For the following exercises, use the table below.

Number of Ballots	8	10	12	4
Option A	1	3	2	1
Option B	3	1	4	4
Option C	4	2	1	2
Option D	2	4	3	3

16. Determine Borda score for each candidate, and the winner of the election using the Borda count method.
 17. Is there a majority candidate? If so, which candidate?
 18. Does the Borda method election violate the majority criterion? Justify your answer.
 19. In a Borda count election, the candidates have the following Borda scores: A 15, B 11, C 12, and D 16. The pairwise match up points for the same voter profiles would have been A 2, B 0, C 1, and D 3. Identify which

fairness criteria, if any, are violated by characteristics of the described voter profile in this Borda election. Explain your reasoning.

20. Determine the winner of the election using the ranked-choice method.
21. If the four voters in the last column rank C ahead of A, which candidate wins by the ranked-choice method?
22. Does this ranked-choice election violate the monotonicity criterion? Explain your reasoning.

For the following exercises, use the table below.

Number of Ballots	15	12	9	3
Option A	1	3	3	2
Option B	2	2	1	1
Option C	3	1	2	3

23. Determine the winner of the election by the Borda method.
24. Does this Borda method election violate the IIA? Why or why not?
25. Which of the ranked voting methods in this chapter, if any, meets the majority criterion, the head-to-head criterion, the monotonicity criterion, and the irrelevant alternatives criterion?

Standard Divisors, Standard Quotas, and the Apportionment Problem

26. Identify the states, the seats, and the state population (the basis for the apportionment) in the given scenario: The reading coach at an elementary school has 52 prizes to distribute to their students as a reward for time spent reading.
27. Use the given information to find the standard divisor to the nearest hundredth. Include the units. The total population is 2,235 automobiles, and the number of seats is 14 warehouses.
28. Use the given information to find the standard quota. Include the units. The state population is eight residents in a unit, and the standard divisor is 1.75 residents per parking space.

Apportionment Methods

29. Which of the four apportionment methods discussed in this section does not use a modified divisor?
30. Determine the Hamilton apportionment for Scenario X in the table below.

	State A	State B	State C	State D	State E	State F	Total Seats
Scenario X	17.63	26.62	10.81	16.01	13.69	15.24	100

31. Does the apportionment resulting from Method X in the table below satisfy the quota rule? Why or why not?

	State A	State B	State C	State D	State E
Standard Quota	1.67	3.33	5.00	6.67	8.33
Apportionment Method X	2	2	5	7	9

For the following exercises, use the table below and the following information: In Wakanda, the domain of the Black Panther, King T'Challa, has six fortress cities. In Wakandan, the word "birnin" means "fortress city." King T'Challa has found 111 Vibranium artifacts that must be distributed among the fortress cities of Wakanda. He has decided to apportion the artifacts based on the number of residents of each birnin.

Fortress Cities	Birnin Djata (D)	Birnin T'Chaka (T)	Birnin Zana (Z)	Birnin S'Yan (S)	Birnin Bashenga (B)	Birnin Azzaria (A)	Total
Residents	26,000	57,000	27,000	18,000	64,000	45,000	237,000
Standard Quota	12.18	26.70	12.65	8.43	29.98	21.08	111

32. Does the Jefferson method result in an apportionment that satisfies or violates the quota rule in this scenario?
33. Find the modified upper quota for each state using a modified divisor of 2,250. Is the sum of the modified quotas too high, too low, or equal to the house size?
34. Use the Adams method to apportion the artifacts. Determine whether it is necessary to modify the divisor. If so, indicate the value of the modified divisor.
35. Does the Adams method result in an apportionment that satisfies or violates the quota rule in this scenario?
36. Use the Webster method to apportion the artifacts. Determine whether it is necessary to modify the divisor. If so, indicate the value of the modified divisor.
37. Does the Webster method result in an apportionment that satisfies or violates the quota rule in this scenario?
38. Which of the four methods of apportionment from this section are the residents of Birnin S'Yan likely to prefer? Justify your answer.
39. Does the change from a standard divisor to a modified divisor tend to change the number of seats for larger or smaller states more?
40. Which of the four apportionment methods—Jefferson, Adams, Hamilton, or Webster—satisfies the quota rule?

Fairness in Apportionment Methods

41. A city purchased five new firetrucks and apportioned them among the existing fire stations. Although your neighborhood fire station has the same proportion of the city's firetrucks as before the new ones were purchased, it now has one fewer. Is this scenario an example of a quota rule violation, the Alabama paradox, the population paradox, the new-states paradox, or none of these?
42. When the number of seats changed from 25 to 26, the standard quotas changed from A 2.21, B 5.25, C 11.27, and D 6.27 to A 2.30, B 5.46, C 11.72, and D 6.52.
 - a. How did the increase in seats impact the apportionment?
 - b. Is this apportionment an example of a paradox? Justify your answer.
43. The school resources officers in a county were reapportioned based on the most recent census. The number of students at Chapel Run Elementary went up while the number of students at Panther Trail Elementary went down, but Chapel Run now has 1 fewer resources officers while Panther Trail has one more than it did previously. Is this scenario an example of a quota rule violation, the Alabama paradox, the population paradox, the new-states paradox, or none of these?

For the following exercises, the house size is 24 seats. When the population of A increases by 28 percent, B increases by 26 percent, and C increases by 15 percent, the standard quotas change from A 3.38, B 6.32, and C 14.30 to A 3.63, B 6.67, and C 13.71.

44. How did the change in populations impact the apportionment?
45. Is this apportionment an example of a paradox? Justify your answer.
46. When the city of Cocoa annexed an adjacent unincorporated community, the number of seats on the city council was increased to maintain the standard ratio of citizens to seats, but one existing community of Cocoa still lost a seat on the city council to another existing community of Cocoa when the new community was added. Is this scenario an example of a quota rule violation, the Alabama paradox, the population paradox, the new-states paradox, or none of these?

For the following exercises, the house size was 27. There were three states with standard quotas of A 6.39, B 11.40, and C 9.21. A fourth state was annexed, and the house size was increased to 35. The new standard quotas are A 6.38, B 11.37, C 9.19, and D 8.06.

47. How did the additional state impact the apportionment?
48. Is this apportionment an example of a paradox? Justify your answer.

For the following exercises, suppose 11 seats are apportioned to States A, B, and C with populations of 50, 129, and 181

people, respectively. Then the populations of States A, B, and C change to 57, 151, and 208, respectively.

49. Demonstrate that the population paradox occurs when the Hamilton method is used.
50. Demonstrate that the population paradox does not occur when the Jefferson method is used. Justify your answer.
51. Demonstrate that the population paradox does not occur when the Adams method is used. Justify your answer.
52. Demonstrate that the population paradox does not occur when the Webster method is used. Justify your answer.

Chapter Test

For the following exercises, use the table below.

Number of Ballots	12	17	15	13
Option A	1	2	4	3
Option B	3	1	3	2
Option C	4	3	1	4
Option D	2	4	2	1

1. Determine the winner of the election by plurality.
2. Determine the Borda scores for each candidate to determine the winner by Borda count method.
3. Create and analyze a pairwise comparison matrix based on the preference summary to determine the winner of the election by pairwise comparison.
4. From the table below use ranked-choice voting to determine the winner of the election.

Number of Ballots	28	5	30	5	16	16
Option L	3	2	1	1	2	3
Option R	1	1	3	2	3	2
Option E	2	3	2	3	1	1

For the following exercises, identify which fairness criteria, if any, are violated by characteristics of the described voter profile. Explain your reasoning

5. In a Borda count election, the candidates have the following Borda scores: A 1345, B 1260, C 685. Candidate B received 51% of the first-place rankings.
6. In a plurality election, the candidates have the following percentages of first place votes: A 25, B 21, C 30, D 24. The pairwise matchup points for the same voter profiles would have been A 3, B 0, C 2, D 2.

For the following exercises, use the table below.

Number of Ballots	13	14	11	12
Option A	2	1	3	3
Option B	3	2	4	1
Option C	4	4	1	2
Option D	1	3	3	4

7. Determine the winner by ranked-choice voting if two of the voters in the second column up-rank the original

- winner. Refer to Question 4. Which fairness criterion, if any, is violated?
8. Determine the winner by ranked-choice voting if candidate R is removed from the election. Refer to Question 4. Which fairness criterion, if any, is violated?

For the following exercises, use this information: The incorporated town of Orange Grove consists of two subdivisions: The Oaks with 1,254 residents, and The Villages with 10,746 residents. A council with 100 members supervises the municipality's operations with representation proportionate to the number of residents.

9. Identify the states, the seats, and the state population (the basis for apportionment) in the given scenario.
10. Determine the standard divisor for the apportionment
11. Determine each state's standard quota rounded to two decimal places.

For the following exercises, use this information: Air Force administration wanted to distribute 27 aircraft across 6 bases based on the number of qualified pilots stationed at those bases. The standard quota is 2.2963. The standard quotas for each base are listed in the table below.

Air Force Base	(A) Alpha	(B) Bravo	(C) Charlie	(D) Delta	(E) Echo	(F) Foxtrot
Pilots	13	12	5	16	7	9
Standard Quota	5.66	5.23	2.18	6.97	3.05	3.92

12. Determine the states' lower quotas and the states' upper quotas.
13. Use Adams's method to apportion the aircraft.
14. Use Jefferson's method to apportion the aircraft.
15. The apportionment of 616 schools to 5 Hawaiian counties by various methods is displayed in the table below.

County	Hawaii	Honolulu	Kalawao	Kauai	Maui
Lower Quota	87	424	0	31	72
Upper Quota	88	425	1	32	73
Jefferson	87	425	1	31	72
Adams	88	422	1	32	73
Webster	87	424	1	31	73

Apportionment by which methods, if any, fail to satisfy the quota rule? Explain your reasoning.

For the following exercises, use this information: The incorporated town of Orange Grove consists of two subdivisions: The Oaks with 1,254 residents, and The Villages with 10,746 residents. A council with 100 members supervises the municipality's operations. The Hamilton method was used to apportion the council seats. The Oaks has 10 seats on the council, while The Villages has 90 seats. The council votes to annex an unincorporated subdivision called The Lakes with a population of 630. They plan to increase the size of the council to maintain the ratio of seats to residents such that the new council will have 100 seats plus the number of seats given to The Lakes.

16. What is the standard divisor from the original apportionment?
17. What is the new house size?
18. Use the Hamilton method to reapportion the seats.
19. Is the reapportionment an example of the new-states paradox? If so, how?

For the following exercises, use this information: determine whether the reapportionment violates the Alabama paradox, the population paradox, or neither. Justify your answer.

20. States A, B, C, and D received 21, 25, 26, and 28 seats respectively. When the population remains the same, but house size is increased, the reapportionment is A 20, B 26, C 27, and D 29.
21. States A, B, C, and D received 21, 25, 26, and 28 seats respectively. When the house size remains the same, the population of state A increased, the population of state B decreased, and the populations of states C and D

remained the same, the reapportionment is A 20, B 26, C 26, and D 28.